

PRACTICE EXAM 4 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 4

The final paper. Sit this close to your exam date.

QUESTIONS 20 WEIGHTING 30/26/24/20 PASS MARK 14 of 20

Application servers in private subnets must download operating system patches from the internet and read objects from an S3 bucket in the same Region.

Nothing on the internet may initiate a connection to them, and the S3 traffic must not traverse the internet.

Which combination meets these requirements? (Choose two.)

- ☐ A A NAT gateway in a public subnet with a route from the private subnets
- ☐ B A gateway VPC endpoint for S3 in the private subnets' route tables
- ☐ C An internet gateway route added to the private subnets
- ☐ D Public IP addresses assigned to the application servers

◆ PAUSE AND DECIDE

Which combination meets these requirements? (Choose two.)

- A

 NAT gateway
Allows outbound connections for patching while blocking anything inbound, which is exactly the patching requirement.
- B

 Gateway endpoint for S3
Keeps S3 traffic on the AWS network rather than routing it out through the NAT gateway, and it is free.
- C

 Internet gateway route
Would make the subnets public, allowing inbound connections. It fails the stated requirement outright.
- D

 Public IP addresses
Makes the servers directly reachable, which the question forbids.

THE SIGNAL

Two separate requirements, two separate answers. Outbound-only means NAT; private S3 access means a gateway endpoint.

A security team must be notified whenever an IAM policy granting full administrative access is attached to any principal in any of forty accounts.

They also want a record of who attached it and when.

Which combination meets both requirements? (Choose two.)

- ☐ A AWS Config with a rule detecting overly permissive policies, aggregated organisation-wide
- ☐ B AWS CloudTrail with an organisation trail delivering to a central S3 bucket
- ☐ C Amazon Macie scanning IAM policy documents
- ☐ D VPC Flow Logs with a metric filter

◆ PAUSE AND DECIDE

Which combination meets both requirements? (Choose two.)

- A

 Config rule, aggregated
Config evaluates resource configuration against rules and notifies, and an aggregator covers all forty accounts.
- B

 Organisation CloudTrail
Records the API call that attached the policy, giving the identity and timestamp the team needs.
- C

 Macie on IAM policies
Macie discovers sensitive data inside S3 objects. It does not evaluate IAM policies.
- D

 Flow Logs
Record network traffic. They say nothing about IAM policy changes.

THE SIGNAL

Config evaluates and notifies; CloudTrail supplies who and when. The pair again, in a new costume.

A partner company must be blocked from a specific range of IP addresses that has been sending malicious traffic to a subnet, regardless of any security group configuration.

The block must apply to every instance in that subnet.

Which control should be used?

- ☐ A A security group rule denying the IP range
- ☐ B A network ACL rule denying the IP range on the subnet
- ☐ C An IAM policy condition on the source IP
- ☐ D A route table entry blackholing the IP range

◆ PAUSE AND DECIDE

Which control should be used?

- A

Security group deny rule
Security groups support allow rules only. There is no way to write a deny in one.
- B

NACL deny rule
NACLs support deny rules and apply at the subnet boundary, covering every instance regardless of its security group.
- C

IAM policy condition
Controls access to AWS APIs, not network traffic reaching an instance.
- D

Route table blackhole
Route tables direct outbound traffic by destination. They do not filter inbound traffic by source.

THE SIGNAL

Blocking a specific address is the one thing only a NACL can do. Security groups cannot deny.

A company wants CloudFront to serve content from an S3 bucket while ensuring the bucket itself is never publicly accessible and can only be read by the distribution.

Content must be served through the company's own domain name over HTTPS.

Which combination should be configured? (Choose two.)

- ☐ A Origin Access Control on the distribution, with a bucket policy permitting only that distribution
- ☐ B An ACM public certificate for the custom domain, associated with the distribution
- ☐ C S3 static website hosting with public read access
- ☐ D A signed URL generated for every object request

◆ PAUSE AND DECIDE

Which combination should be configured? (Choose two.)

- A** Origin Access Control
Lets CloudFront read a private bucket, with a bucket policy that permits the distribution and nothing else.
- B** ACM certificate for the domain
Provides HTTPS on the custom domain, free with CloudFront and renewed automatically.
- C** Static website hosting
Requires public read access, which directly violates the stated requirement.
- D** Signed URLs per object
Restricts which users may access content. It does not control how CloudFront reaches the bucket.

THE SIGNAL

OAC solves bucket privacy. ACM solves the custom domain over HTTPS. Two requirements, two features.

An organisation must ensure that CloudTrail logging cannot be stopped or deleted by anyone in any of its accounts, including local account administrators.

The security team currently receives an alert when logging stops and re-enables it by hand.

Which solution meets the requirement?

- ☐ A An AWS Config rule with automatic remediation in each account
- ☐ B A service control policy denying CloudTrail stop and delete actions
- ☐ C An IAM policy attached to every principal denying those actions
- ☐ D A CloudWatch alarm notifying the security team faster

◆ PAUSE AND DECIDE

Which solution meets the requirement?

- A

 Config with remediation
Detective. Logging stops and then resumes, leaving a gap. 'Cannot be stopped' is not satisfied by re-enabling afterwards.
- B

 SCP denying the actions
Preventive, and it sits above account administrators so the API call fails outright and cannot be removed locally.
- C

 IAM policy on principals
Right idea, wrong scope. A local administrator can detach it, and it needs maintaining in every account.
- D

 Faster alarm
Notification only. It speeds up the current process rather than preventing the action.

THE SIGNAL

'Cannot' means preventive. Detection and remediation answer a different question.

A financial services firm requires that encryption keys be generated and stored in single-tenant hardware validated to FIPS 140-3 Level 3, under the firm's exclusive control.

Regulation does not permit shared key infrastructure.

Which service should be used?

- ☐ A AWS KMS with a customer managed key
- ☐ B AWS CloudHSM
- ☐ C AWS KMS with imported key material
- ☐ D AWS Secrets Manager with automatic rotation

◆ PAUSE AND DECIDE

Which service should be used?

- A

 KMS customer managed key
Gives you policy control, but KMS hardware is multi-tenant, which the regulation forbids.
- B

 CloudHSM
Dedicated single-tenant hardware security modules under your exclusive control, meeting the stated FIPS validation level.
- C

 KMS with imported material
Lets you originate the key material, but it still resides in shared KMS infrastructure.
- D

 Secrets Manager
Stores and rotates secrets. It is not a key management hardware service.

THE SIGNAL

CloudHSM has one signal: dedicated single-tenant hardware or a specific FIPS validation level.

A photo-sharing application must generate a thumbnail, scan for prohibited content, and update a search index whenever an image is uploaded. Three separate teams own these. The scanning service is offline for maintenance most Sunday nights, and no upload event may be lost.

Which architecture meets these requirements?

- ☐ A S3 events to an SNS topic with the three services subscribed directly
- ☐ B S3 events to an SNS topic with three SQS queues subscribed
- ☐ C S3 events to a single SQS queue read by all three services
- ☐ D The upload service calling each of the three in sequence

◆ PAUSE AND DECIDE

Which architecture meets these requirements?

- A

 SNS direct subscriptions
SNS does not store. Events published while the scanning service is offline on Sunday nights are lost.
- C

 One shared queue
A message is consumed once, so whichever service reads it first prevents the other two from seeing it.
- B

 SNS fan-out to three queues
Each consumer has its own durable buffer, so the offline scanner loses nothing and the others are unaffected.
- D

 Sequential direct calls
Tightly coupled. The offline scanner blocks the upload, and each new consumer requires a producer change.

THE SIGNAL

The regular maintenance window is what upgrades SNS alone to SNS plus SQS.

A business requires that a Region failure cause no more than a few seconds of data loss and no more than a few minutes of downtime for a relational workload.

Budget has been approved for continuously running infrastructure in a second Region.

Which approach is MOST appropriate?

- ☐ A Backup and restore from cross-Region snapshots
- ☐ B Pilot light with core services switched off
- ☐ C Warm standby with an Aurora global database
- ☐ D Nightly database dumps copied to the second Region

◆ PAUSE AND DECIDE

Which approach is MOST appropriate?

- A

 Backup and restore
Recovery in hours. It cannot meet a few minutes of downtime.
- B

 Pilot light
Meets the data loss objective but recovery takes tens of minutes while services start, which exceeds a few minutes.
- C

 Warm standby with Aurora global
A running scaled-down environment recovers in minutes, and Aurora global replication is typically under a second.
- D

 Nightly dumps
Risks losing up to twenty-four hours, far outside a seconds-level RPO.

THE SIGNAL

A few minutes of RTO is tighter than pilot light delivers. The approved budget confirms warm standby.

An application uses an Auto Scaling group behind an Application Load Balancer across two Availability Zones. During a zone impairment, the surviving zone's instances became overwhelmed and the service degraded.

The group's minimum is four instances, two per zone, and scaling took several minutes to react.

Which change would MOST improve resilience?

- | | |
|---|--|
| ☐ A Spread across three Availability Zones and size capacity to survive losing one | ☐ B Increase the maximum capacity of the existing group |
| ☐ C Add a second Application Load Balancer | ☐ D Move to a single larger instance per zone |

◆ PAUSE AND DECIDE

Which change would MOST improve resilience?

- A

 Three AZs, sized for N minus one
Losing one of three zones removes a third of capacity rather than half, and provisioning for that loss removes the reliance on slow scaling.
- B

 Raise the maximum
Scaling still takes several minutes, so the service degrades during the gap regardless of the ceiling.
- C

 Second load balancer
The load balancer was not the failure. It is already zone-redundant.
- D

 One larger instance per zone
Reduces redundancy. Losing one zone still removes half the capacity.

THE SIGNAL

Across two zones, losing one costs fifty percent. Across three, it costs thirty-three. Size for the loss rather than relying on scaling.

A company must keep a copy of critical S3 data in a second Region for disaster recovery, and must be able to prove to auditors that every object has been replicated.

Objects are added continuously throughout the day.

Which solution meets these requirements?

- ☐ A S3 Cross-Region Replication with versioning enabled and replication metrics
- ☐ B A nightly AWS DataSync task copying the bucket
- ☐ C A lifecycle policy transitioning objects to the second Region
- ☐ D S3 Transfer Acceleration to the destination bucket

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

CRR with versioning and metrics

Replicates continuously as objects arrive, requires versioning on both buckets, and replication metrics provide the audit evidence.
- B

Nightly DataSync

Leaves up to a day unreplicated, and objects arrive continuously throughout the day.
- C

Lifecycle transition

Lifecycle rules move objects between storage classes, not between Regions.
- D

Transfer Acceleration

Speeds uploads from distant locations. It is not a replication mechanism.

THE SIGNAL

Versioning is a prerequisite for cross-Region replication. Option C describes something lifecycle rules cannot do.

An order service writes to a queue consumed by a fulfilment service. A recent deployment bug caused malformed messages that the consumer cannot parse. Those messages are retried endlessly, blocking valid orders behind them.

The team needs valid orders to keep flowing and wants to inspect the failures afterwards.

What should be configured?

- ☐ A A dead-letter queue with a maximum receive count
- ☐ B A longer message retention period
- ☐ C A FIFO queue to guarantee ordering
- ☐ D A shorter visibility timeout

◆ PAUSE AND DECIDE

What should be configured?

- A

Dead-letter queue
After the configured number of failed attempts the poison message moves aside, unblocking valid work and preserving the failures for inspection.
- B

Longer retention
Keeps the malformed messages around longer, which makes the blockage last longer.
- C

FIFO queue
Strict ordering would make the blockage worse, since nothing behind a stuck message could proceed.
- D

Shorter visibility timeout
Makes failed messages reappear sooner, increasing the retry rate rather than stopping it.

THE SIGNAL

Poison messages blocking a queue is precisely what a dead-letter queue exists to solve.

A company must run forty thousand independent analysis jobs each week. Jobs take between fifteen minutes and eight hours, some need GPU instances, and some must wait for an earlier job to finish.

Custom scripts currently manage queuing and provisioning, and frequently leave instances running after jobs complete.

Which service is designed for this workload?

- ☐ A AWS Batch
- ☐ B Amazon EMR
- ☐ C AWS Lambda with SQS
- ☐ D AWS Step Functions driving EC2 instances

◆ PAUSE AND DECIDE

Which service is designed for this workload?

- A

AWS Batch
Job queues, dependencies between jobs, GPU support, and automatic provisioning and termination of compute. Every requirement matches.
- B

Amazon EMR
Runs distributed frameworks like Spark, where one job spans a cluster. These are independent jobs.
- C

Lambda with SQS
Jobs run up to eight hours, far past the fifteen-minute ceiling, and Lambda offers no GPU.
- D

Step Functions with EC2
Orchestrates workflows but does not manage a job queue or provision and terminate batch compute.

THE SIGNAL

'Independent' separates Batch from EMR. Job queues plus dependencies plus automatic compute means Batch.

Clickstream events arrive at forty thousand records per second and must be written to S3 in Parquet format. A delay of a minute or two is acceptable.

The team will not write or operate stream-processing code.

Which service should be used?

- ☐ A Kinesis Data Streams with a Lambda consumer
- ☐ B Kinesis Data Firehose with format conversion
- ☐ C Amazon MSK with a Kafka Connect S3 sink
- ☐ D SQS with a Lambda function batching writes

◆ PAUSE AND DECIDE

Which service should be used?

- A

 Data Streams with Lambda
Works, but you own the consumer code, batching, retries and shard management, which the question excludes.
- B

 Firehose with format conversion
Fully managed delivery, native Parquet conversion, and buffering that fits the one to two minute window. No code at all.
- C

 MSK with Kafka Connect
The most operational work of any option, and Kafka is not a stated requirement.
- D

 SQS with Lambda
SQS distributes work rather than carrying a high-throughput ordered stream, and you would still write the batching.

THE SIGNAL

Near-real-time plus no code means Firehose. Sub-second or replay would mean Data Streams.

An on-premises imaging application must read and write to a local file share for performance, while the authoritative copy of the data lives in S3 and is used by cloud analytics.

The site cannot hold the full archive locally, but staff must open recent files immediately.

Which solution meets these requirements?

- | | |
|---|---|
| ☐ A AWS DataSync on an hourly schedule | ☐ B AWS Storage Gateway in File Gateway mode |
| ☐ C Amazon EFS mounted over a VPN | ☐ D AWS Snowball Edge kept permanently on site |

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

 DataSync hourly
Copies data on a schedule. It does not present a local share, and the site cannot hold the full archive.
- C

 EFS over VPN
Every read crosses the network, failing the local performance requirement.
- B

 File Gateway
Presents a local file share backed by S3, caching recently used data on site while the full archive stays in the cloud.
- D

 Snowball on site
A transfer device, not permanent hybrid storage, with no ongoing S3 integration.

THE SIGNAL

DataSync transfers. Storage Gateway provides ongoing local access. That sentence separates them.

A logistics company must move ninety terabytes of archives to S3. The site has a 200 Mbps internet connection shared with daily operations and it cannot be saturated.

The migration must finish within three weeks.

Which approach should be used?

- ☐ A AWS DataSync over the existing connection
- ☐ B AWS Snowball Edge devices
- ☐ C S3 Transfer Acceleration with multipart upload
- ☐ D Order a Direct Connect circuit and transfer over it

◆ PAUSE AND DECIDE

Which approach should be used?

- A

 DataSync over the internet
Ninety terabytes at 200 Mbps takes over forty days at full line rate, and the line cannot even be fully used.
- C

 Transfer Acceleration
Optimises use of available bandwidth. It cannot create bandwidth that does not exist.
- B

 Snowball Edge
Physical transfer sidesteps the bandwidth constraint entirely and fits comfortably inside three weeks.
- D

 Direct Connect
Provisioning takes weeks to months, so the circuit may not exist before the deadline, for a one-off move.

THE SIGNAL

Do the division before reading the options. Volume against line speed against deadline decides it.

Twelve EC2 instances must share a Windows file system with Active Directory integration and SMB access.

The application is a legacy Windows product that cannot be modified.

Which storage service should be used?

- ☐ A Amazon EFS
- ☐ B Amazon FSx for Windows File Server
- ☐ C Amazon FSx for Lustre
- ☐ D Amazon S3 with a mounted file interface

◆ PAUSE AND DECIDE

Which storage service should be used?

- A

 Amazon EFS
An NFS file system for Linux. It does not provide SMB or Active Directory integration.
- C

 FSx for Lustre
A high-performance file system for HPC and machine learning, not Windows file sharing.
- B

 FSx for Windows File Server
Native SMB shares with Active Directory integration, which is exactly what an unmodifiable Windows application expects.
- D

 S3 with a file interface
Object storage. A legacy Windows application expecting an SMB share will not work against it.

THE SIGNAL

Windows plus SMB plus Active Directory is FSx for Windows. Linux shared files is EFS.

A public web application runs an Auto Scaling group between ten and eighty instances. Ten are required at all times and the rest scale with unpredictable demand.

The application is stateless and instances can be replaced at any time. The company wants lower cost with the service continuously available.

Which approach is MOST cost-effective?

- ☐ A Run all instances on Spot
- ☐ B Cover the ten baseline instances with a Savings Plan and use Spot above that
- ☐ C Purchase Reserved Instances for eighty instances
- ☐ D Run everything On-Demand

◆ PAUSE AND DECIDE

Which approach is MOST cost-effective?

- A

 Everything on Spot
A simultaneous reclamation of the pool could take the whole service down. The baseline must be guaranteed.
- B

 Savings Plan baseline plus Spot
The always-on ten get a commitment discount, the variable surge runs at Spot prices, and the service survives Spot reclamation.
- C

 Reserve eighty
Pays continuously for peak capacity that is rarely used. Reserve the floor, never the ceiling.
- D

 All On-Demand
Takes no discount at all on ten instances running every hour of the year.

THE SIGNAL

Baseline plus variable means commitment pricing for the floor and Spot for the surge.

A company must retain seismic survey data for thirty years to satisfy regulation. Less than half a percent is ever retrieved, and when it is, a wait of up to a day is acceptable. The archive is one and a half petabytes and the priority is the lowest possible storage cost.

Which storage class should be used?

- ☐ A S3 Glacier Flexible Retrieval
- ☐ B S3 Glacier Deep Archive
- ☐ C S3 Standard-IA
- ☐ D S3 Glacier Instant Retrieval

◆ PAUSE AND DECIDE

Which storage class should be used?

- A

 Glacier Flexible Retrieval
Faster retrieval than needed at a higher storage price. The question allows a full day.
- C

 Standard-IA
Immediate access nobody needs, at many times the storage cost over thirty years.
- B

 Glacier Deep Archive
The cheapest storage AWS offers, and its twelve-hour retrieval fits comfortably inside the one-day tolerance.
- D

 Glacier Instant Retrieval
Millisecond access for data retrieved a few times a year. Far more than this requirement asks for.

THE SIGNAL

Match retrieval speed to the stated tolerance. Paying for faster than asked is a wrong answer.

A finance team cannot explain why the S3 bill has risen steadily across eighty buckets in sixteen accounts. There is no inventory of what each bucket holds or how it is accessed. They want to understand where the cost is concentrated before changing anything.

What should the solutions architect recommend?

- ☐ A Apply Intelligent-Tiering to every bucket immediately
- ☐ B Enable S3 Storage Lens with the advanced tier across the organisation
- ☐ C Set a lifecycle policy on every bucket transitioning to Glacier at thirty days
- ☐ D Enable versioning to track object changes

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Intelligent-Tiering everywhere
A defensible later step, but applied blindly it adds monitoring charges to buckets that may simply need deleting.
- C

 Blanket Glacier lifecycle
Would break buckets needing immediate access and archives data that might be better expired.
- B

 S3 Storage Lens
Gives organisation-wide visibility of storage, growth and access patterns, which is exactly what the team asked for.
- D

 Versioning
Stores every previous version, which increases the bill rather than explaining it.

THE SIGNAL

The question asks how to find out, not what to change. Measure before acting.

A company's monthly bill rose by fourteen thousand dollars with no known change. Leadership wants to know which service, account and resource caused it, wants automatic notification if it recurs, and wants raw line-item data for their own analysts. They have no existing cost tooling configured.

Which combination meets these requirements? (Choose two.)

- ☐ A Cost Explorer to investigate, plus Cost Anomaly Detection for future alerts
- ☐ B The Cost and Usage Report delivered to S3 and queried with Athena
- ☐ C An AWS Budget set at last month's total
- ☐ D AWS Config recording resource changes

◆ PAUSE AND DECIDE

Which combination meets these requirements? (Choose two.)

- A

 Cost Explorer plus Anomaly Detection
Cost Explorer finds where the increase came from, and Anomaly Detection alerts automatically on unusual future spend.
- B

 CUR queried with Athena
The most granular billing data available, which is what independent analyst investigation requires.
- C

 Budget at last month's total
Alerts on a fixed threshold, which misses gradual growth and identifies no specific resource.
- D

 AWS Config
Records configuration changes, not cost. It attributes no spending to anything.

THE SIGNAL

Cost Explorer investigates, Anomaly Detection alerts on the unusual, CUR is the raw data. Three jobs, three tools.

Reading the trend across four papers

The shape of your four scores tells you more than the last one.

- ◆ **Rising, finishing at 16 or above.** Book the exam. You are performing above the pass mark with room to spare.
- ◆ **Flat in the mid teens.** Passing is plausible but not comfortable. Target your weakest domain specifically rather than rewatching everything.
- ◆ **Rising but finishing at 13 or 14.** The trend is right and the level is not there yet. Give it another week of focused work.
- ◆ **Erratic, swinging by four or more.** Usually reading speed rather than knowledge. Practise the five-step question routine from module 17.

ON THE EXAM

Practice scores tend to run slightly high, because these questions come from a course you have just watched. Leave yourself a margin.

You do not need to know every service. You need to **read a scenario for its constraint** and choose accordingly.

A final checklist for exam day

- ◆ **Read the last line first** on every question. The qualifier is the criterion.
- ◆ **Never leave a blank.** There is no penalty for a wrong answer.
- ◆ **Flag and move on at ninety seconds.** Aim to finish the first pass in a hundred minutes.
- ◆ **Trust your first reading** unless you can articulate a specific reason to change.
- ◆ **Fifteen questions are unscored.** An unreasonably obscure one may simply not count.

ON THE EXAM

That is the course complete. Eighteen modules, eighty original practice questions, and every scenario written from scratch.